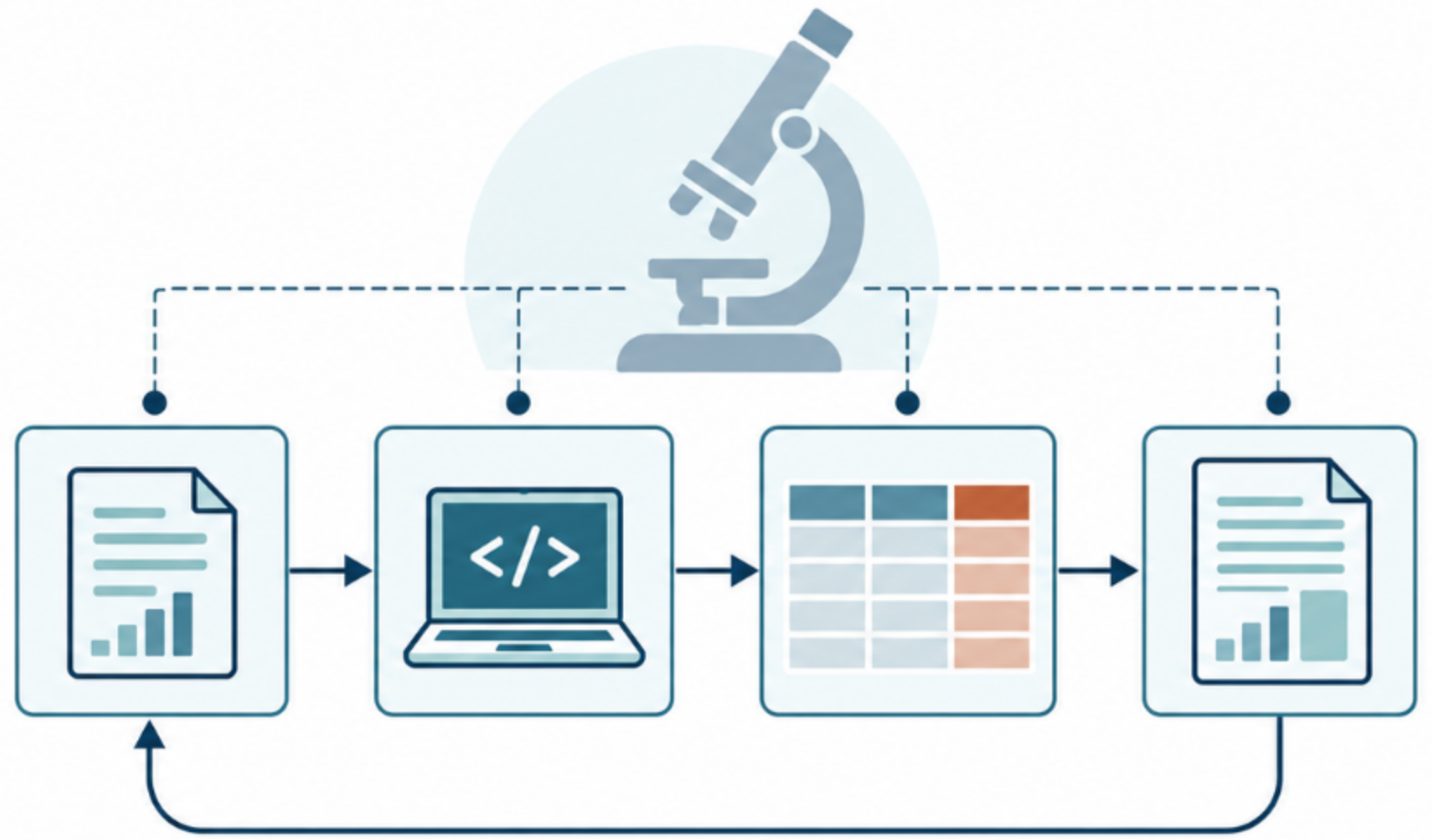


AI for science

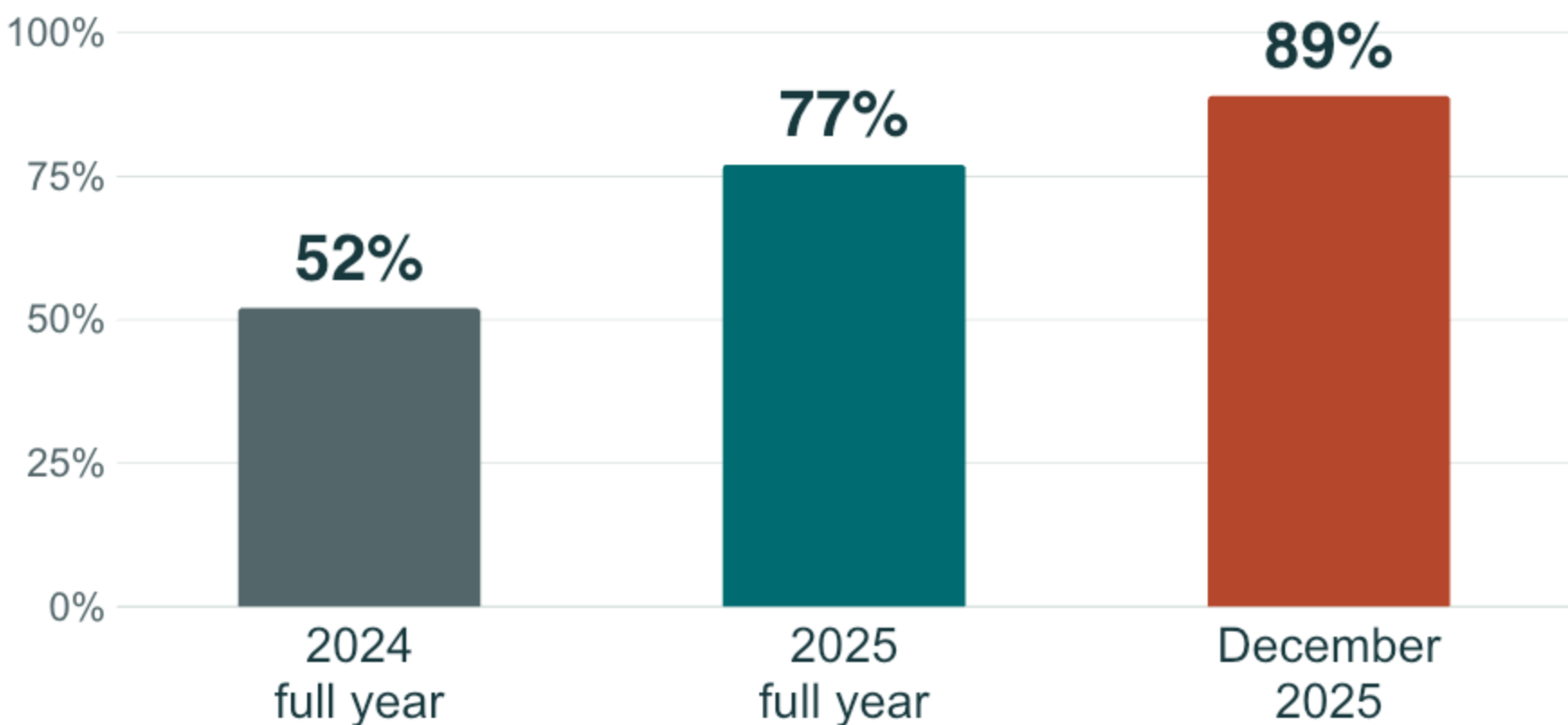
A practical workshop
with Codex desktop

Andres Vallejo Pulido
9 September 2026



MOST BIOMEDICAL PUBLICATIONS SHOW SIGNS OF LLM-ASSISTED WRITING

Nature News · 20 August 2026 · reporting an August preprint



MOST BIOMEDICAL PUBLICATIONS SHOW SIGNS OF LLM-ASSISTED WRITING

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August 12, 2026

Abstract

Over the past several years, LLM-powered chatbots and agents have become widely used as a tool for academic writing. LLM-assisted writing can be valuable by removing language barriers but at the same time causes concerns about misconduct and fraud. To inform policy decisions, it is necessary to monitor the prevalence of LLM-altered texts in scholarly publications. Despite some recent progress in this direction, no existing method can produce reliable estimates. Here we suggest and validate a new unbiased approach to estimate LLM usage in a corpus of texts based on changing word frequencies. We apply our method to the full texts of open-access biomedical papers from Pubmed Central, and show that by the end of 2025, 89% of papers show excess of LLM-associated vocabulary. We also find that LLMs are twice as likely to be used when writing a paragraph in the Discussion section (68%) compared to a paragraph in the Methods section (32%), but even inside the Methods section, the overall prevalence of LLM usage is over 50%. We believe that our estimates are crucial to shape future guidelines and policies.

715v1 [cs.CL] 11 Aug 2026

A vocabulary-based population estimate; it does not establish AI use in a particular paper.

What authors may use AI for in a paper

Elsevier generative AI policy for journals, simplified

1 Manuscript preparation



Allowed to support:

- Literature synthesis and field overviews
- Identifying research gaps and generating ideas
- Content organisation
- Improving language and readability



Human authors must review, verify, edit, and take full responsibility.

2 Explanatory figures



Allowed for non-data illustrations such as:

- Flow charts
- Decision trees
- Timelines
- Conceptual schematics
- Experimental workflow diagrams



Disclose AI use in the image caption and in the manuscript AI statement.

3 Data visualisations



- Allowed only when the figure is directly derived from the underlying data.
- Use reproducible analytical, computational, or statistical methods.



Report the AI tool, version, and developer in the Methods section.

4 AI used in the research methods



- Allowed when AI is part of the actual research workflow, for example image analysis, prediction, or modelling.



Describe the tool and version reproducibly in the Methods section.

5 Disclosure exceptions



No declaration needed for:

- Basic spelling, grammar, and punctuation checks
- Disability-related assistive technology used only for accessibility



Substantive changes to writing or structure should still be disclosed.

Always required



Human oversight



Accuracy checks



Transparency and disclosure



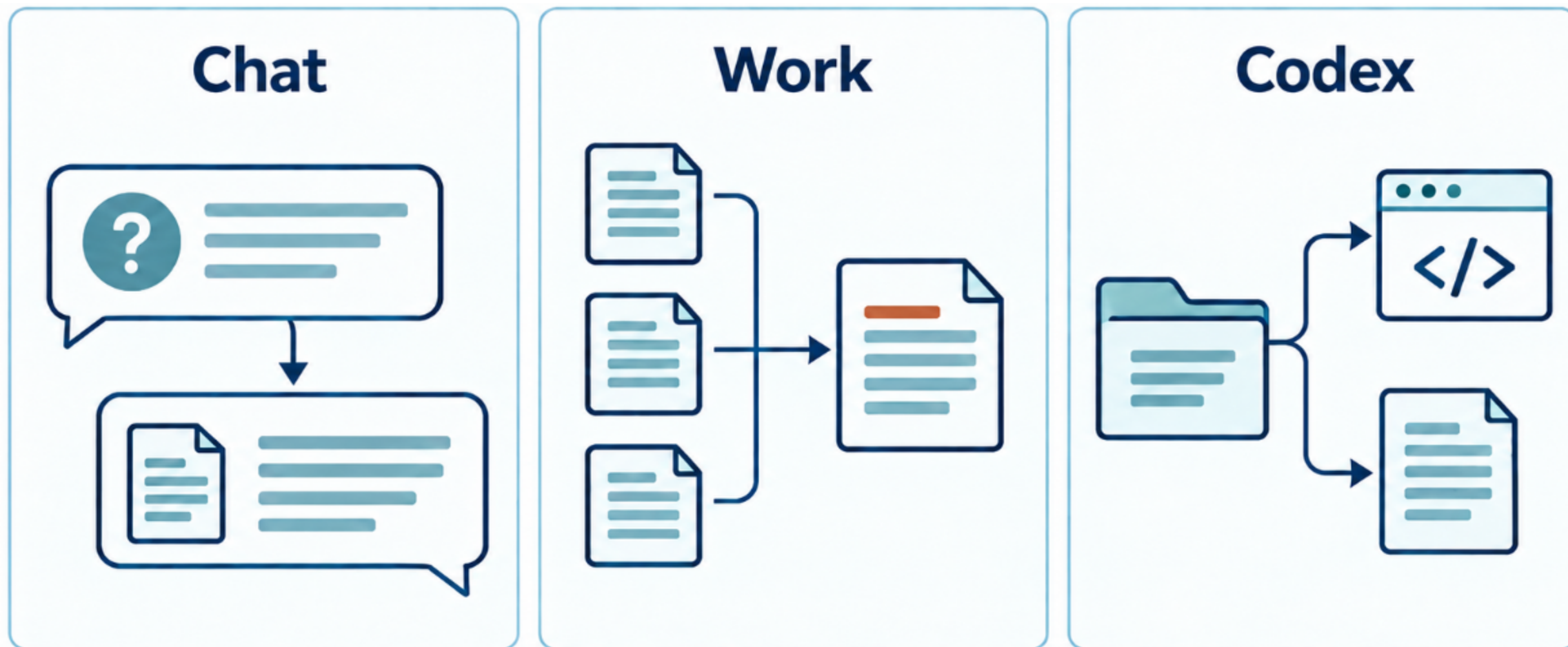
Privacy and confidentiality



AI is not an author

Chat, Work and Codex

The desktop app can contain more than a chat window



Examples: explain a confidence interval · build a briefing · audit project files

Browser chat can use tools too. Access, execution environment and permissions make the difference.

What is an Agent?

An agent is software that can understand a goal, make a plan, use tools, take actions, and check results to complete a task.

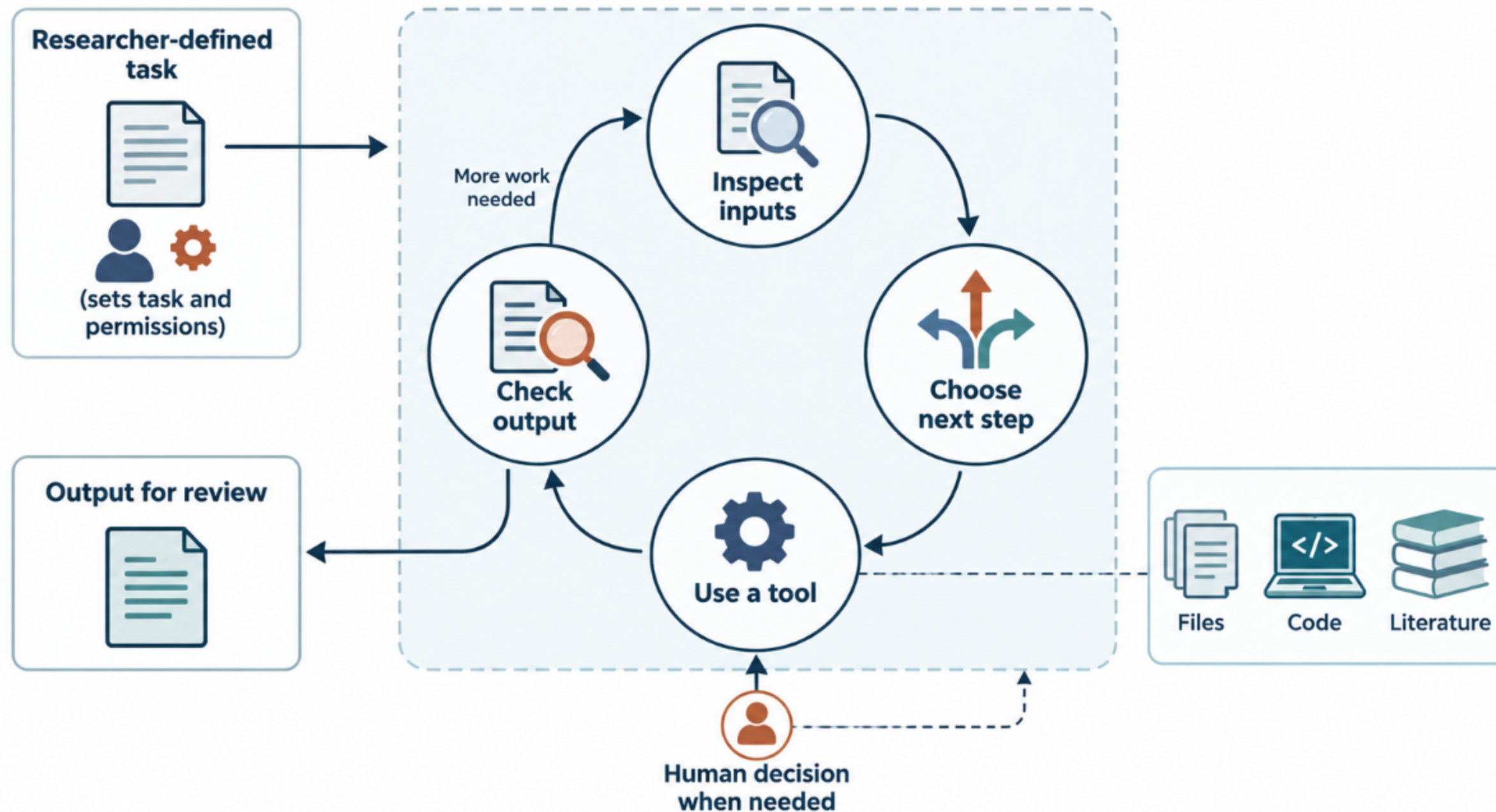


Analogy: like a good personal assistant, an agent does more than answer questions, it takes steps to get the job done.

- ✓ Understands your goal
- ✓ Makes a plan
- ✓ Uses the right tools
- ✓ Takes action
- ✓ Checks results
- ✓ Delivers the outcome

From goals to results

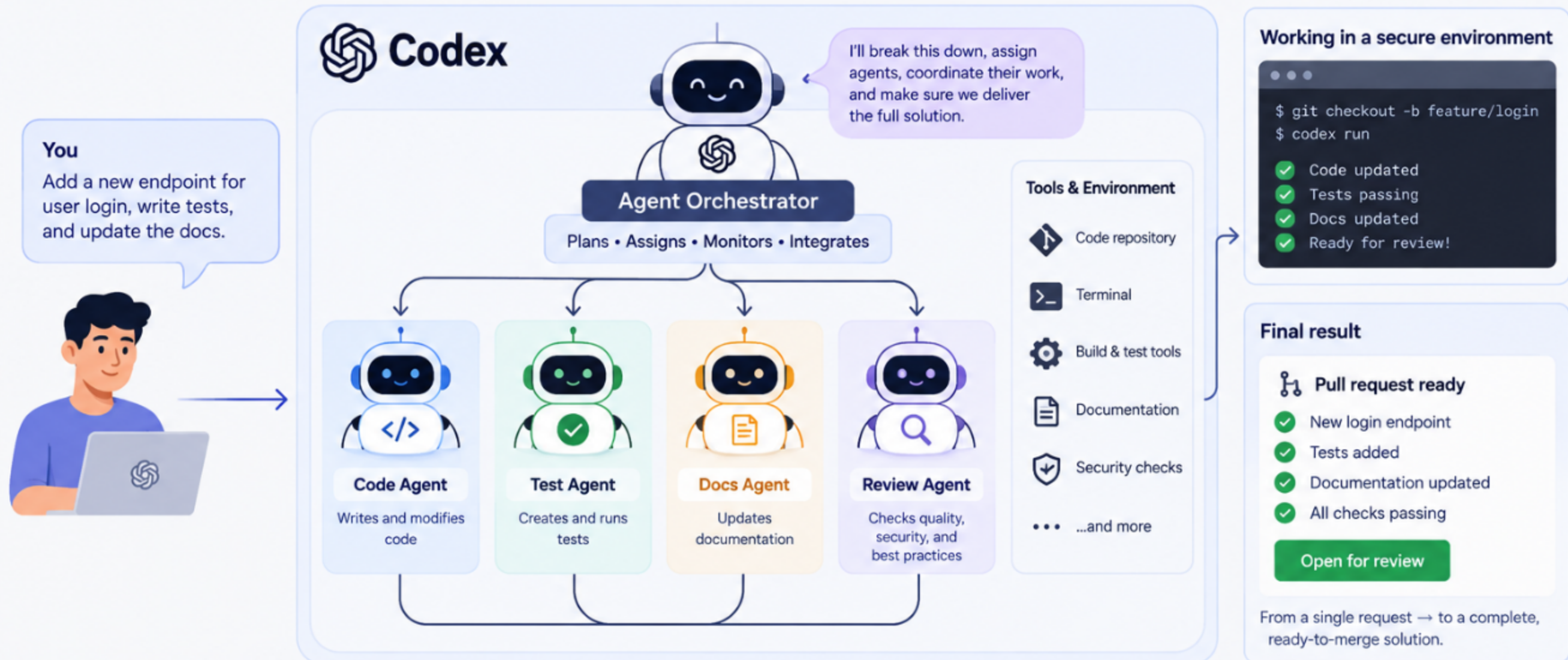
What is an agent?



Inside Codex: Agents in Action

Multiple specialized agents, working together, orchestrated to turn your request into real results.

Codex uses agent orchestration — a team of specialized AI agents that plan, collaborate, and take action in a secure coding environment, just like a real software team.



1

You give a goal

Describe what you want in plain language.

2

Orchestrator plans

Breaks the task down and assigns it to the right agents.

3

Agents take action

Specialized agents work in parallel, using tools, writing code, running tests, updating docs, and more.

4

Results delivered

Codex integrates the work and prepares a ready-to-review result.

Agents + Orchestration = Real progress

Codex turns your ideas into working software, with a team of AI agents.

More builders.
Less busywork.

Scientific tasks worth trying

Choose outputs whose accuracy you can assess



Literature

Build an evidence table

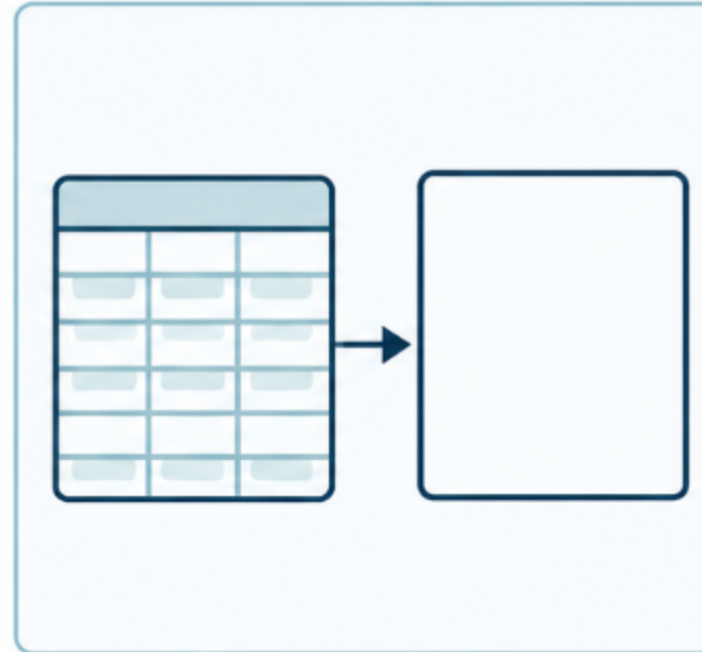
Verify its sources



Data

Audit sample and donor IDs

Expose every mismatch



Figures

Plot the supplied table

Match data and code



Manuscript

Check Results against tables

Check effects and citations

Start with a bounded task. Keep biological interpretation and final decisions under review.

Claude vs ChatGPT: Messi vs Ronaldo

Claude / Messi

ChatGPT / Ronaldo



Choose the fit for today's task.

A playful analogy, not a benchmark or a fixed ranking

What actually differs

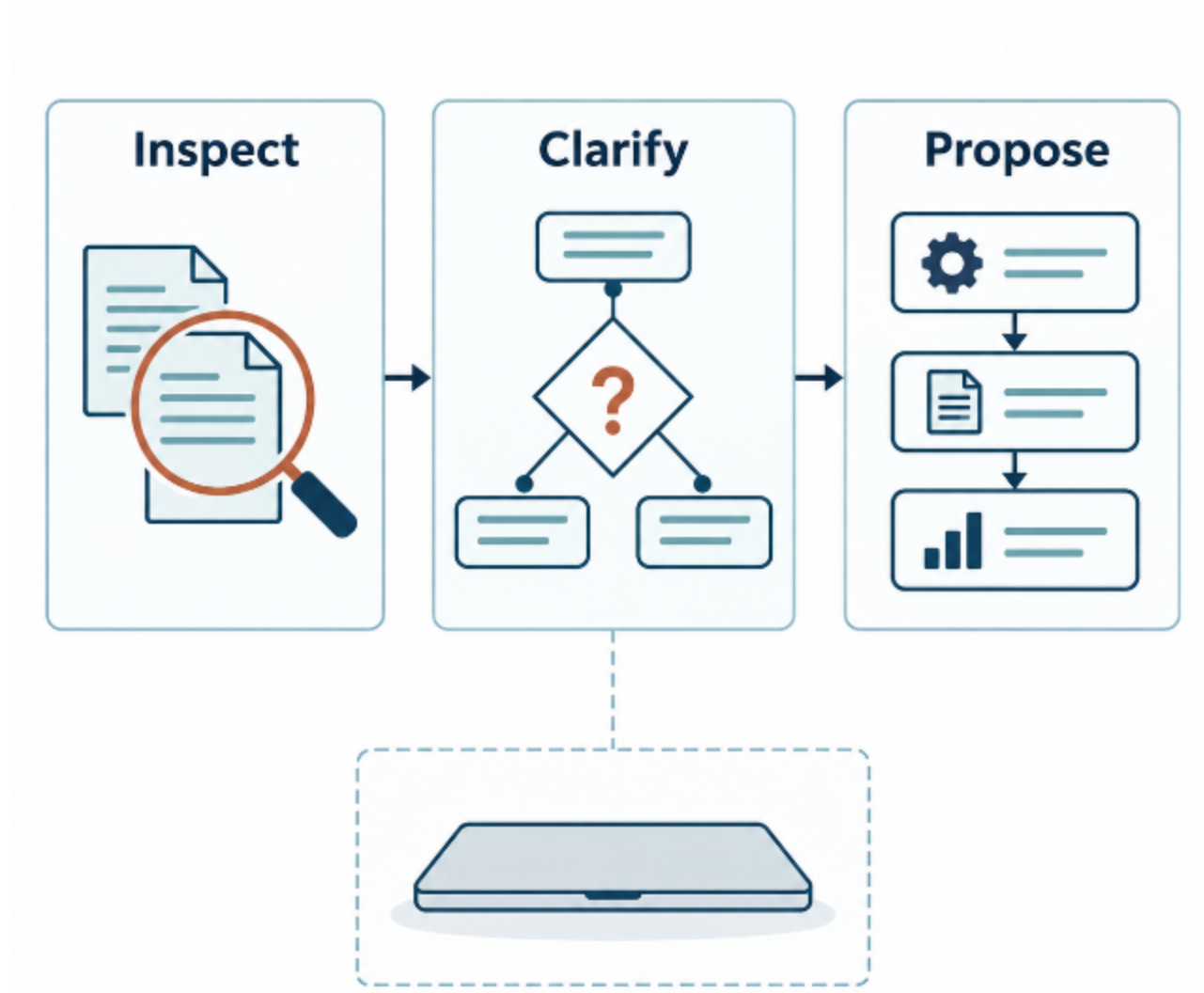
Compare the product and workflow, as well as the model

Workflow	Anthropic / Claude	OpenAI / ChatGPT
Conversation	Claude Chat	ChatGPT Chat
General tasks	Cowork	Work
Project files and code	Claude Code	Codex
Extensions	Skills, plugins, connectors	Skills, plugins, MCP connections

Try the same abstract or sample sheet. Score the checked output, time and effort required.

Plan mode

Agree on the approach before implementation



Inspect

Use read-only tools to establish the inputs.

Clarify

Resolve scientific decisions and missing evidence.

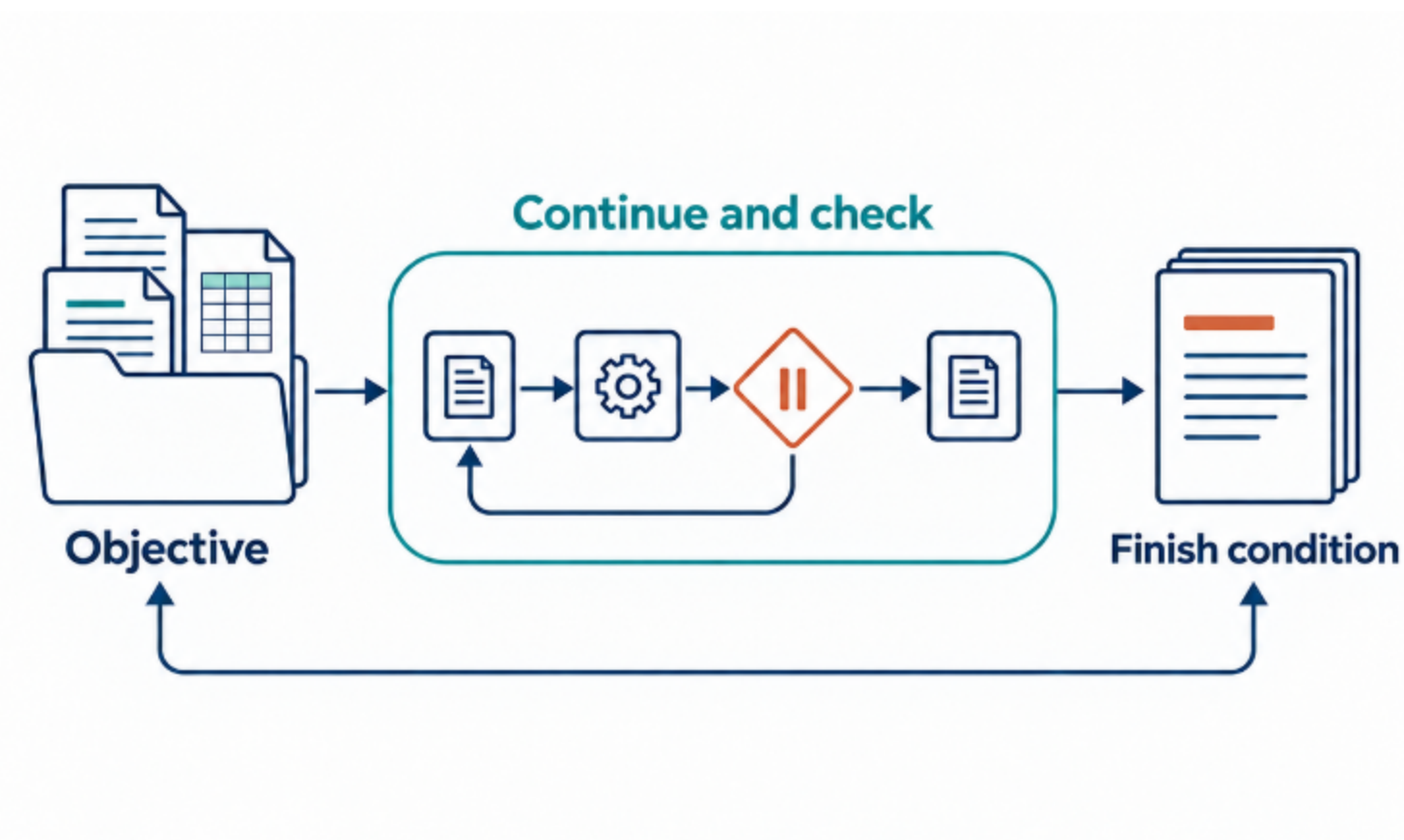
Propose

Review the steps, outputs and checks.

Leave Plan mode when ready to implement. A plan has not executed or validated the analysis.

Goals

Keep working towards a defined, verifiable finish



Objective

Audit every CSV in the folder.

Finish condition

Every input covered; saved outputs checked.

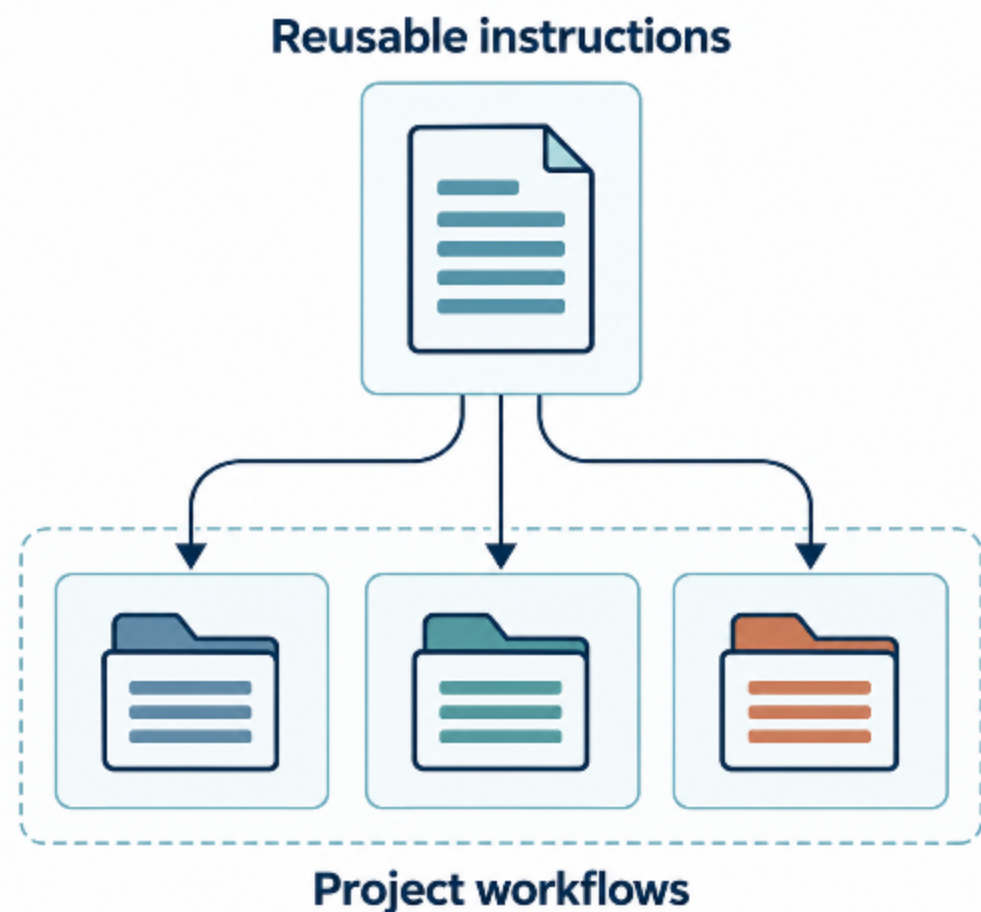
Limits and control

Preserve inputs; resolve decisions; pause or resume where supported.

A goal can continue across turns. It is not a recurring schedule or a guarantee of success.

Skills

Reusable instructions for a task you repeat



A reusable lab protocol

SKILL.md describes when to use the workflow, its steps and checks.

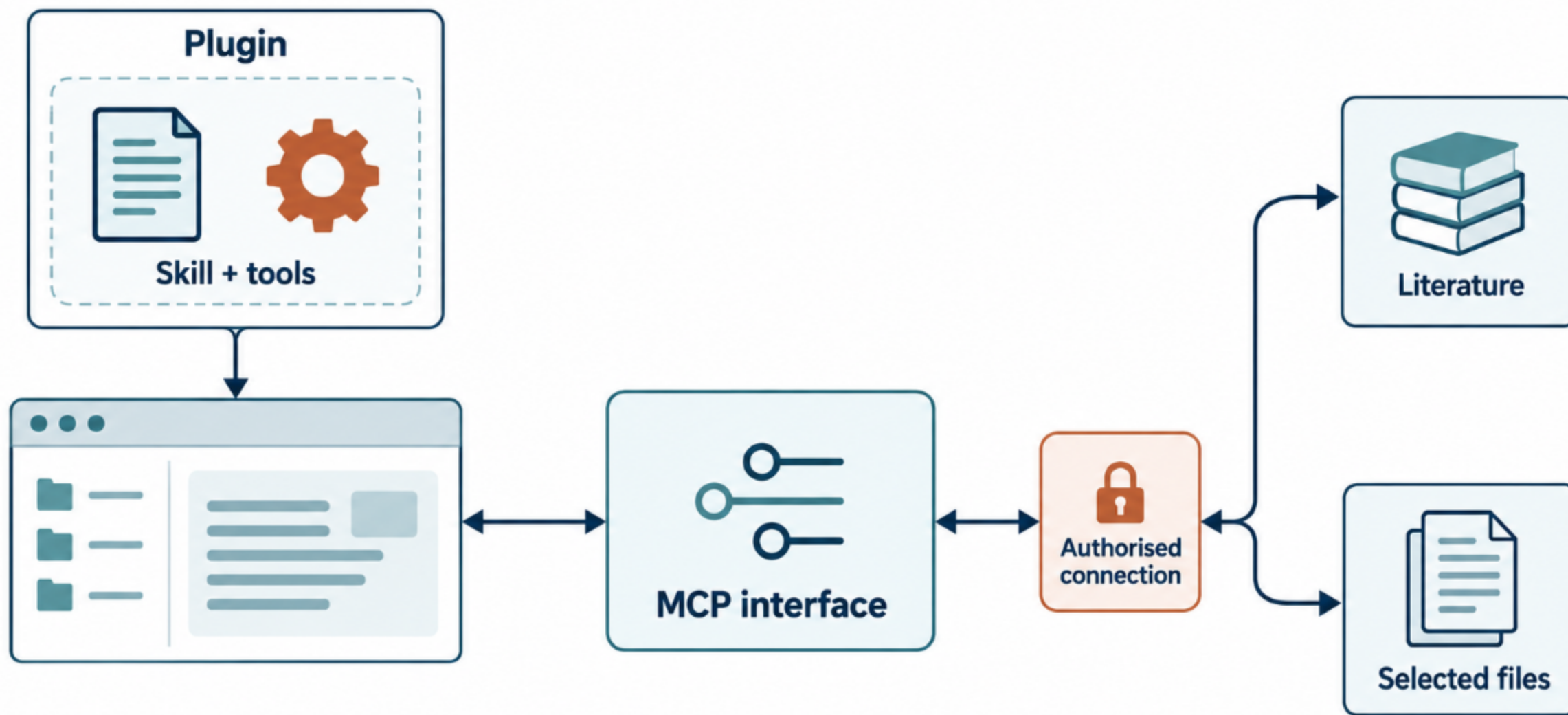
Example: sample-sheet audit

Read → match IDs → report problems
→ save evidence and preserve inputs

May include scripts, references or templates.

A skill guides existing capabilities. It does not train the model or grant access to your data.

Plugins, connections and MCP

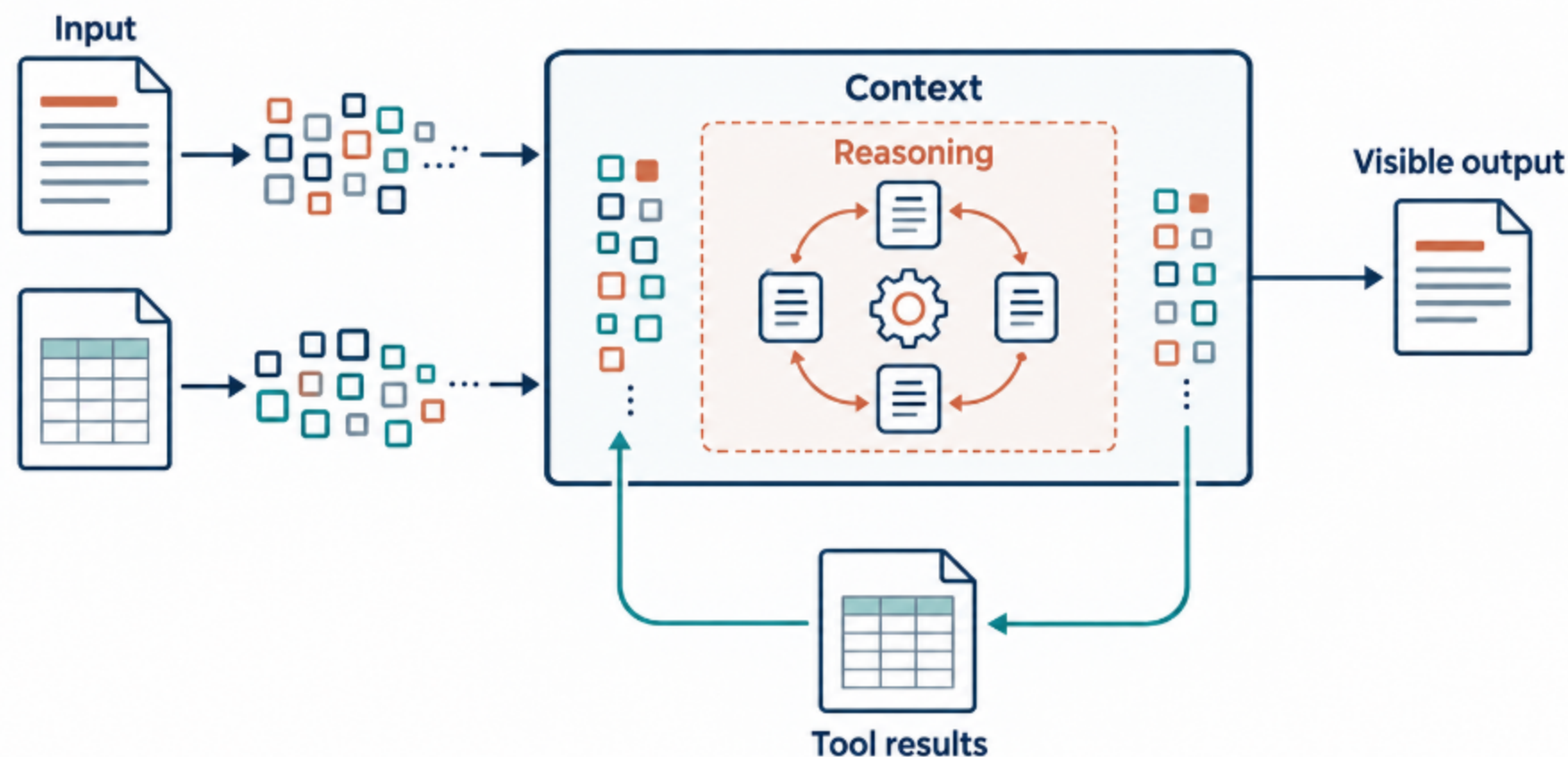


Example of a connected workflow. A skill gives instructions; the connection sets access.

A plugin may need a separate connection. Not every tool uses MCP; access still has limits.

Tokens: what the model processes

A token is a small unit of information, not necessarily a whole word



Input

Prompt, context and tool results

Processing + output

Reasoning plus the visible answer

Repeated calls

An agent can revisit tools and context

Context window: information considered at once. Cached input may receive a lower rate.

A short final answer can still require substantial processing.

Luna, Terra and Astra: usage trade-offs

Illustrative workload: 20,000 input + 5,000 output tokens at Standard speed

Illustrative credits



GPT-5.6 Luna

Repeatable extraction and formatting

GPT-5.6 Terra

Everyday analysis, documents and figures

GPT-6 Astra

Complex reasoning and demanding review

Same illustrative token workload; rates do not predict the cost of a whole task.

Compare the cost of a checked result. Cheaper per token does not guarantee a cheaper completed task.

Example only: uncached input; total output includes reasoning. Other feature charges excluded.

Reasoning effort and Fast mode

Two separate controls beneath the model choice



Reasoning effort



Speed

Reasoning effort

Low / Light → quick tasks

Medium → balanced

High / Extra High → more thought

Max / Ultra → demanding work

Fast mode

2.5x

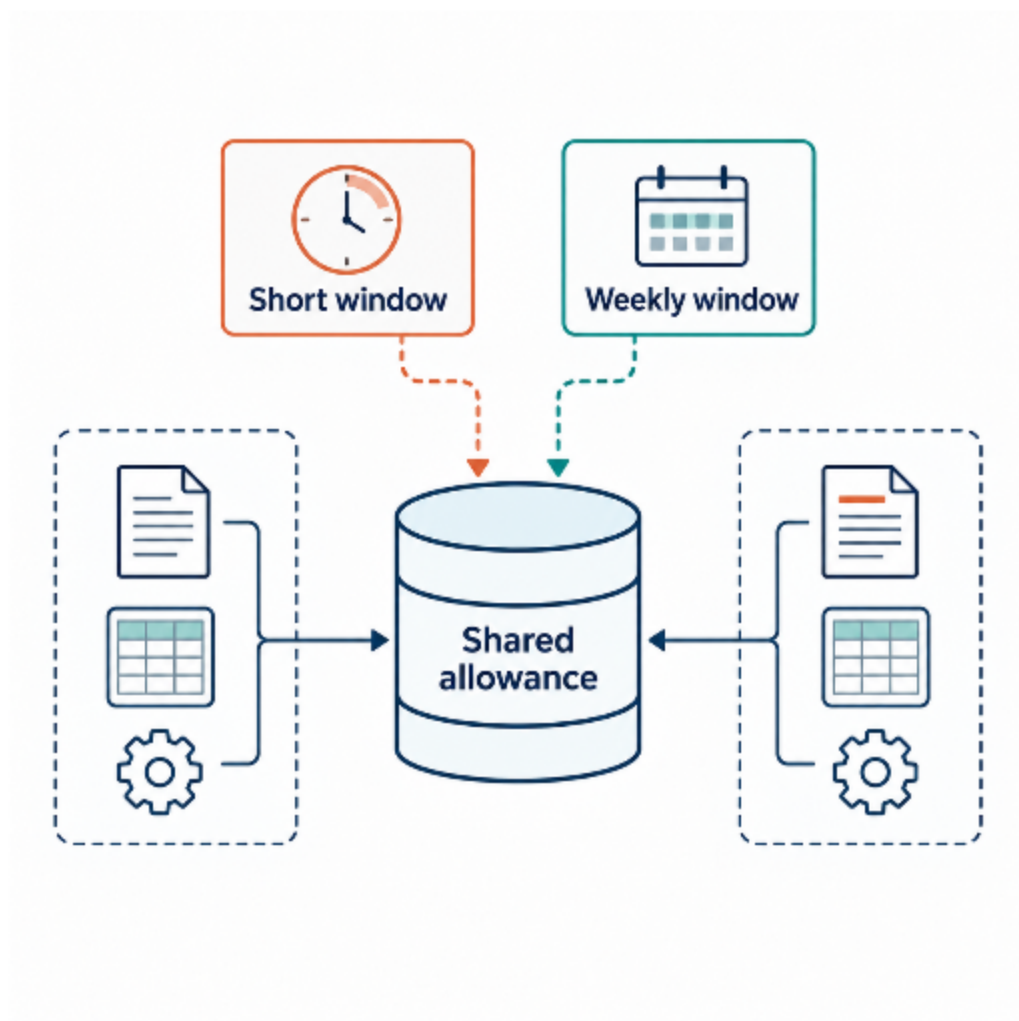
Standard credit rate
Astra, Terra and Luna
with ChatGPT sign-in

Changes speed; reasoning is a separate setting.

Use extra reasoning when the problem needs it, and Fast when lower latency is worth the usage.

Weekly allowance and reset windows

A shared usage budget, not a fixed number of prompts



Shared allowance

Codex and Work draw from the same budget.

Separate windows

Five-hour and weekly limits may apply.

Check the reset

Use the usage view or /status where supported.

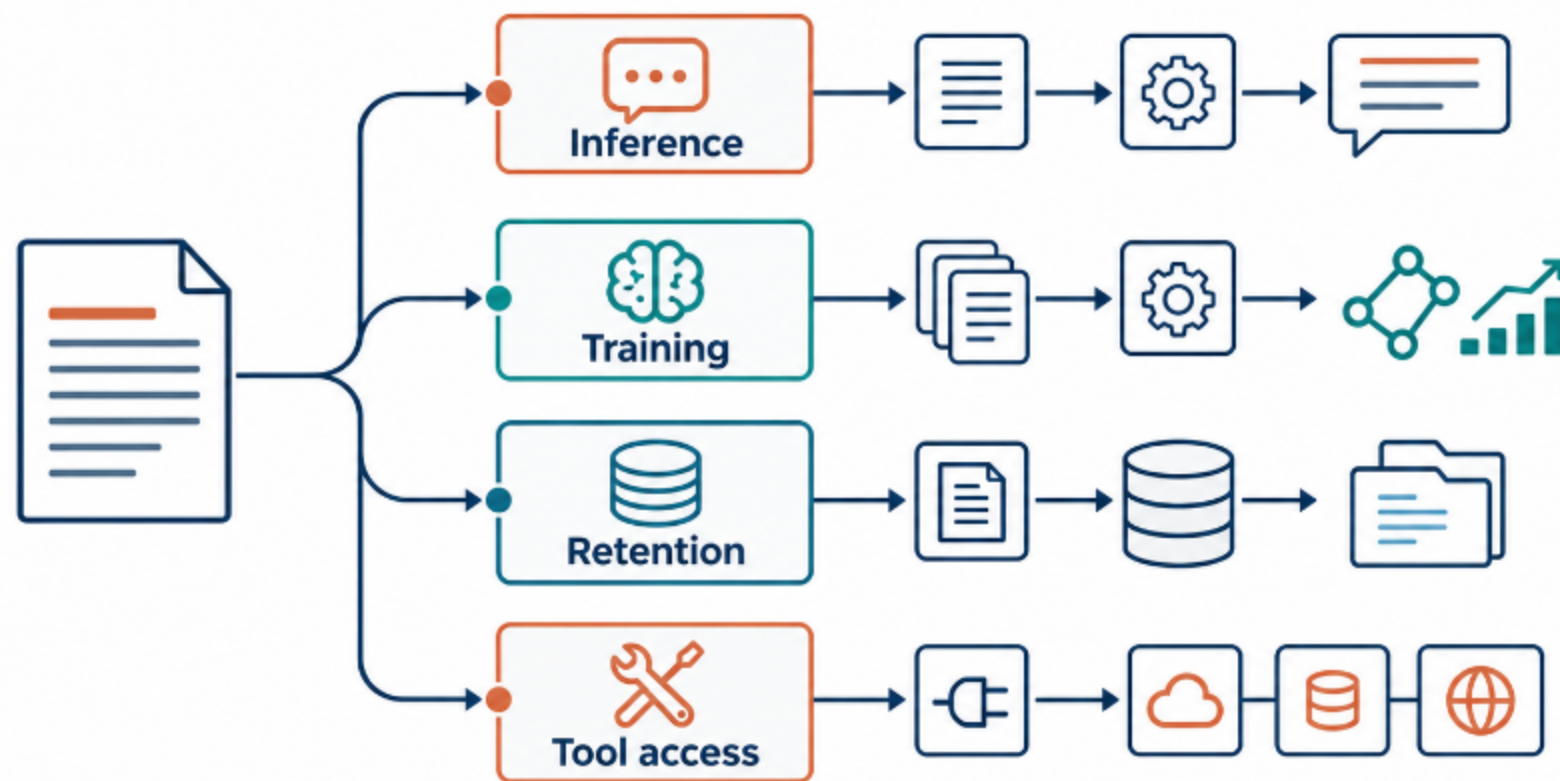
Separate units

Tokens, allowance, credits and API bills differ.

The same-looking task can use different amounts because context, tools and reasoning differ.

Training, retention and access

Four questions to ask before sharing research data



Inference

Where is it processed?

Training

Can it improve future models?

Retention

What is stored, and how long?

Tool access

Which services and files?

Turning training off does not delete chats, saved files or copies held by other services.

ChatGPT accounts and Codex

Public policy overview, checked 9 September 2026



Account / mode	Training	Retention
Personal	May be used; opt-out available. Feedback can be an exception.	Chats retained until deleted; normally removed within 30 days after deletion.*
Temporary Chat	Not used while temporary.	Safety copies: up to 30 days. Saving makes it a regular chat.
Business / Enterprise / Edu	Not used by default.	Storage rules still apply; Enterprise/Edu retention controls.
OpenAI API	Not used by default.	Abuse logs: up to 30 days.* Feature storage varies; approved ZDR.

Codex also has a separate full-environment training control. Review it alongside chat settings.

*Security/legal and de-identification exceptions apply.

Claude accounts and Claude Code

The sign-in route and organisation matter

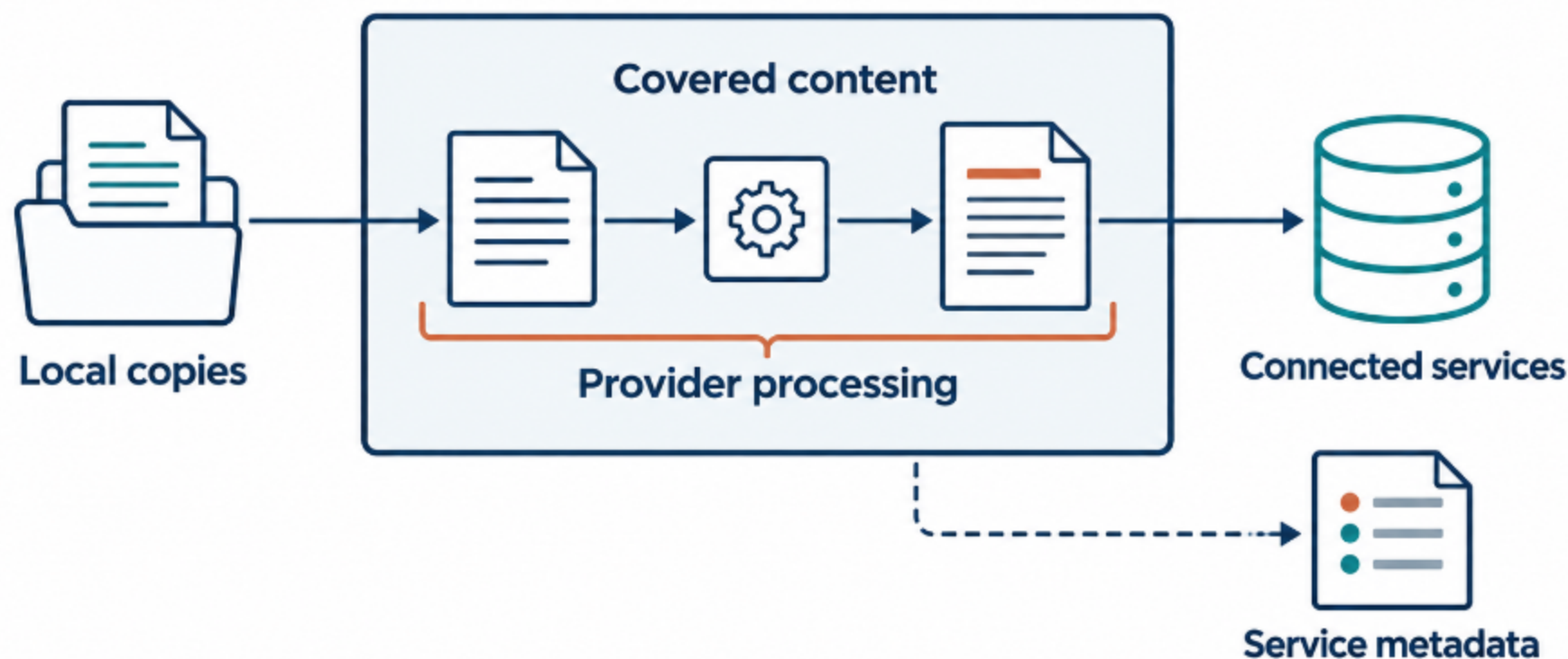


Account / mode	Training	Retention
Personal Free / Pro / Max	Model Improvement consent; feedback also matters.	After deletion: within 30 days.* Consented training copies: up to 5 years.
Incognito	Not used for improvement.	Automatic deletion within 30 days; stated exceptions.
Team / Enterprise	Not used by default; explicit sharing/feedback exceptions.	Chats retained. Enterprise: default indefinite; custom minimum 30 days.
Anthropic API	Not used by default.	Standard deletion within 30 days.* Eligible ZDR arrangements differ.

***Legal, safety and feature exceptions apply. Check covered-model rules and the actual ZDR agreement.**

Zero Data Retention (ZDR)

Coverage depends on the agreement, model and feature



Covered path

Approved organisation, model, endpoint and feature.

Other copies

Local files, service metadata and connected systems.

Exceptions

Safety, legal rules and feature storage affect coverage.

Ask which content, which systems and which exceptions the promise covers.

Paper figures with a verified statistical plan

Synthetic CD8 T-cell activation data

10 donors $\times$ 4 conditions $\times$ 2 technical measurements

Two paired treatment comparisons; no interaction test

Run	Instructions
A	Basic prompt + shared brief and analysis plan
B	Same task + revised scientific-figure skill
C	Same task + revised skill + bounded review loop

Three fresh subagents · same runtime · 183 mm vector figure masters

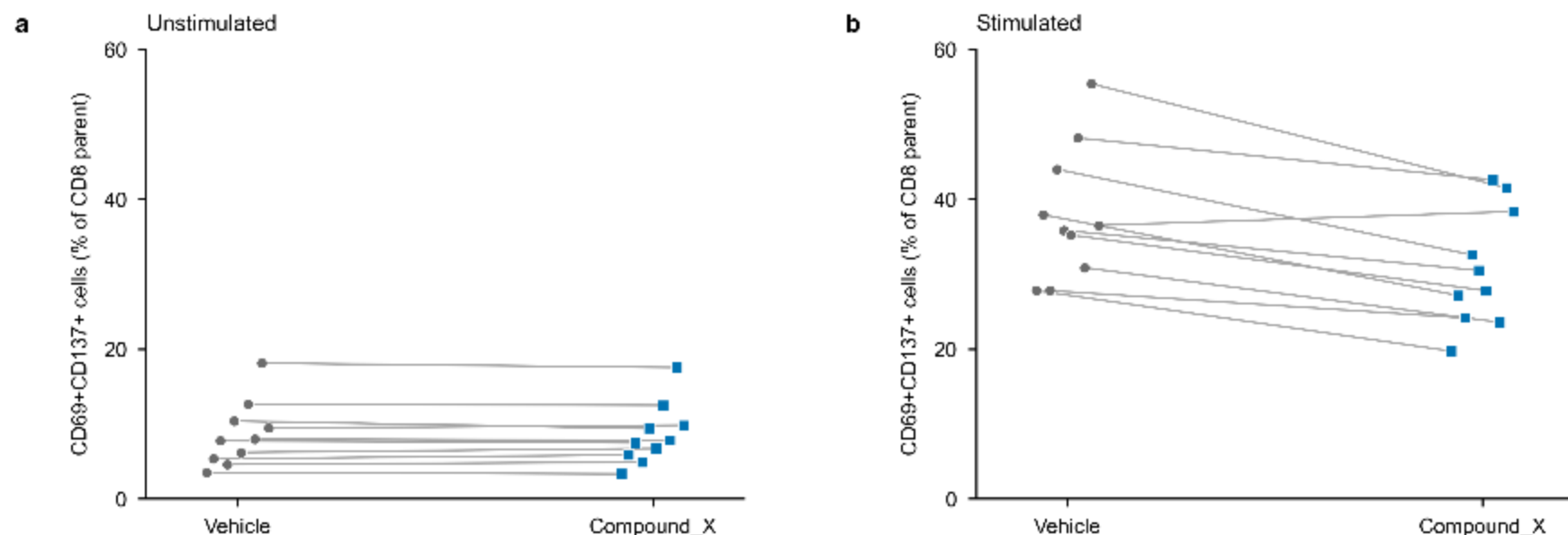
Paired t-tests · Holm correction over two tests · marginal 95% confidence intervals

A · Basic prompt with the common brief

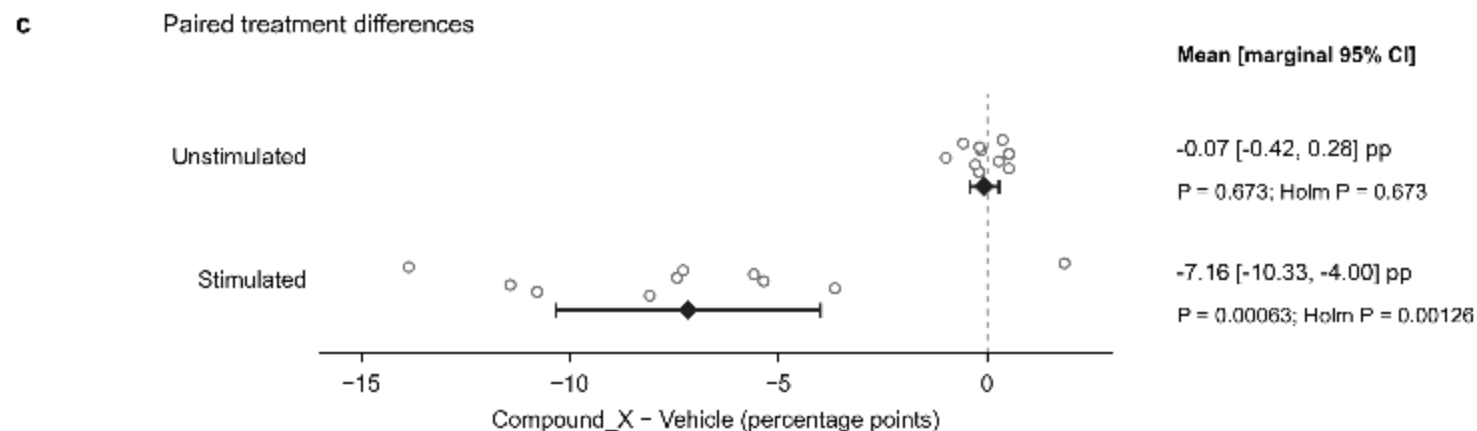
One design version · no supplied figure skill · full production brief

Synthetic CD8 T-cell activation experiment

10 matched donors · 4 cultures per donor · 2 technical aliquots averaged per culture



Each point is one culture mean; lines connect the same donor. Both panels use the same scale.



Open circles: donor differences · Diamonds and bars: mean and marginal 95% CI · Two-sided paired t-tests

Paired donors remain visible.

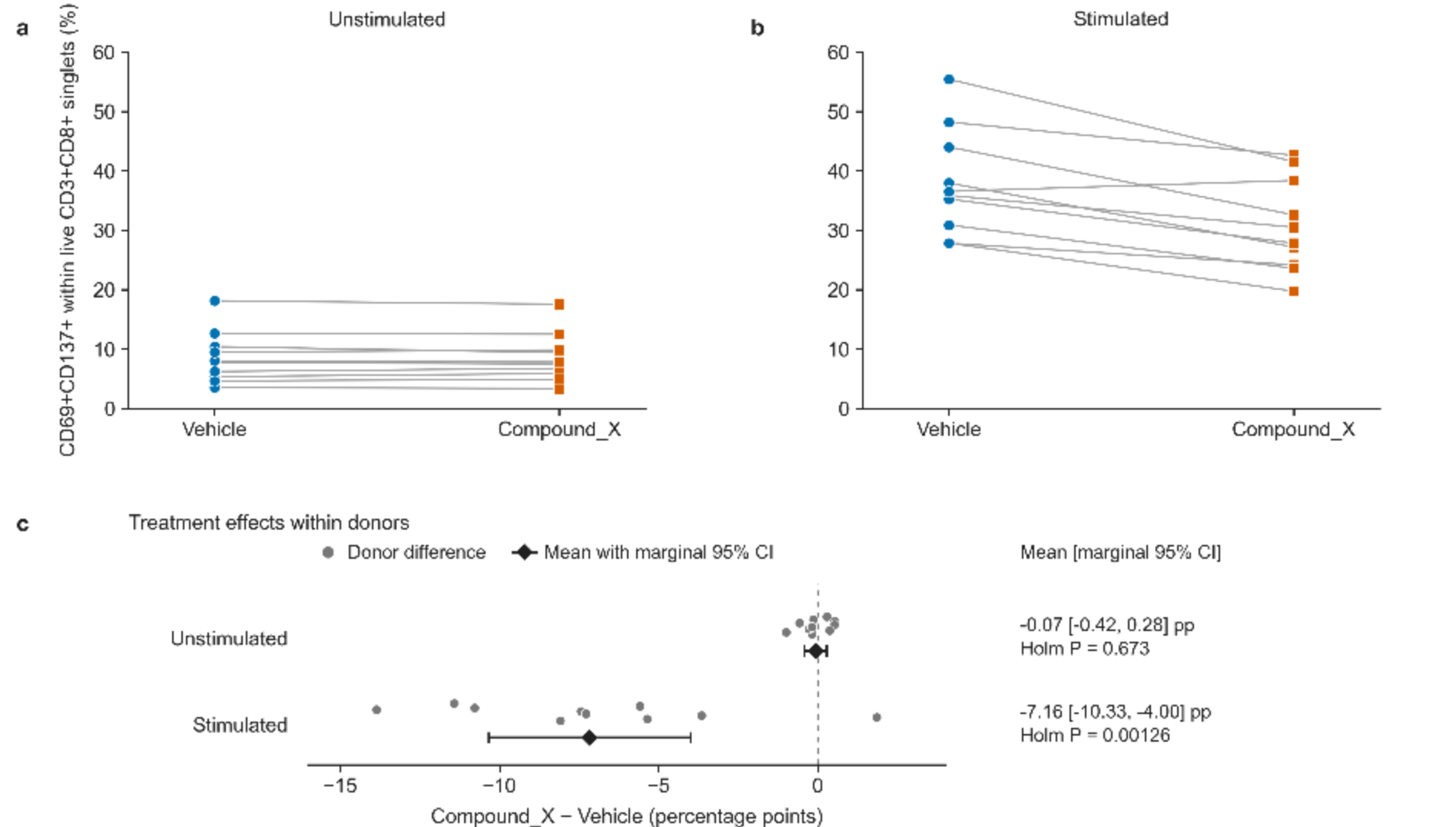
The estimation panel adds mean changes and marginal 95% CIs.

Fresh regeneration passed.

B · Revised skill and basic prompt

A missing symbol was corrected; the first design was retained

Synthetic teaching data



A missing legend glyph was replaced with a vector marker.

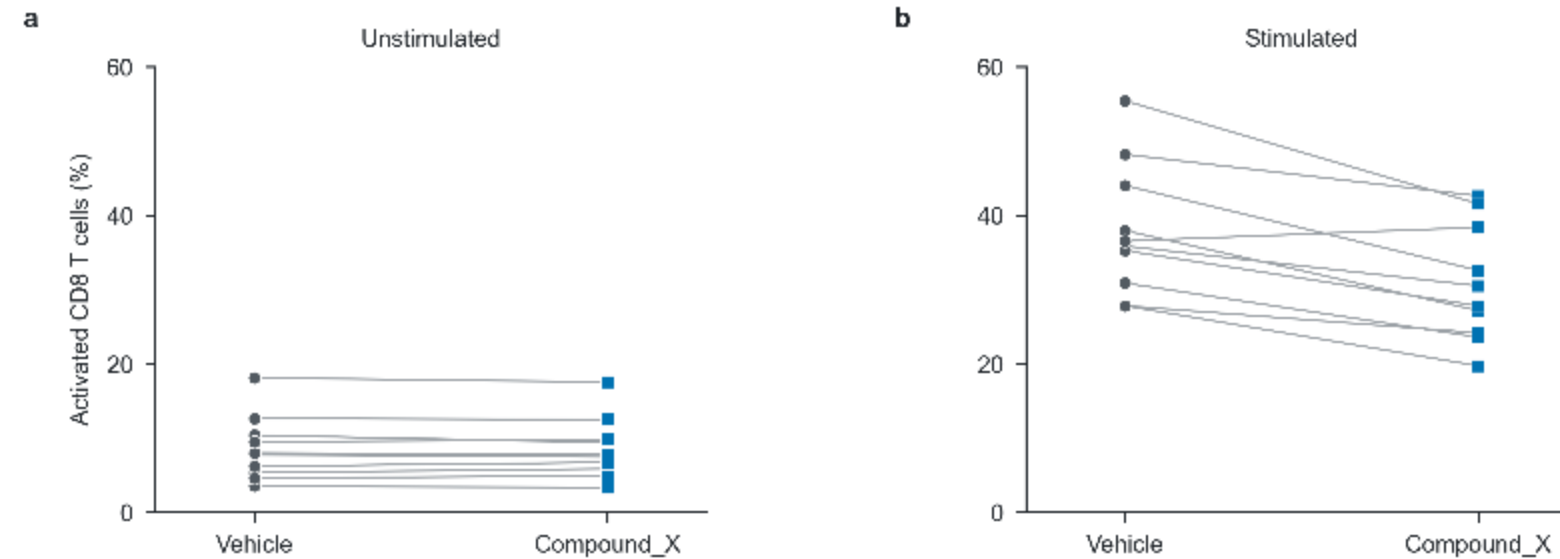
Statistics stayed identical, and fresh regeneration passed.

C · Revised skill and bounded review loop

Initial design retained · zero revisions · version archive and review log

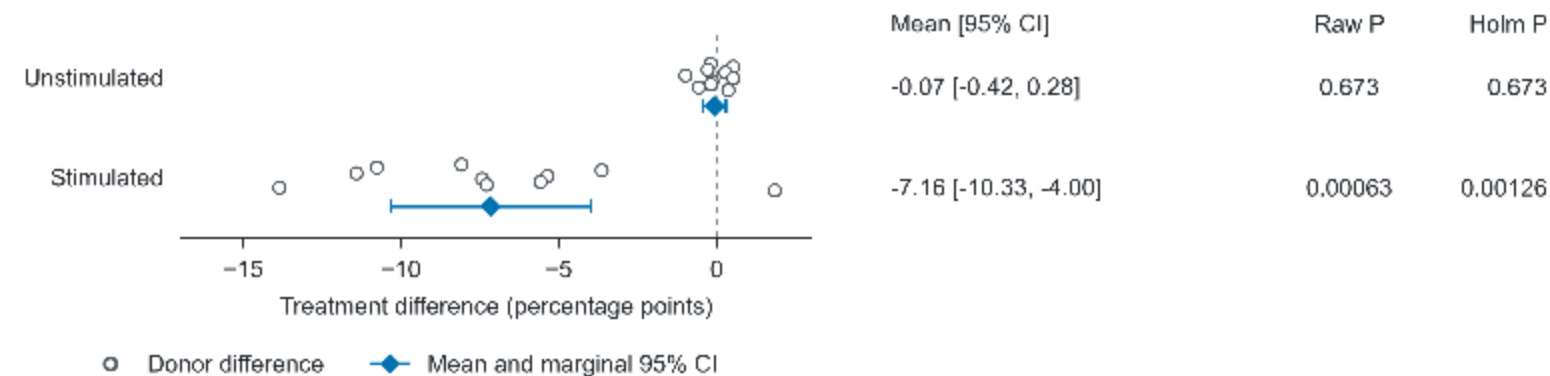
Synthetic CD8 T-cell activation

10 donors | 2 technical aliquots per culture



Points: culture means; lines: matched donors. Same donors in both conditions.

c Treatment effects Compound_X minus Vehicle



The effect panel aligns estimates, confidence intervals and P values.

The explicit loop stopped after v1.

Fresh regeneration passed.

Scientific skills: reusable workflows

Scientific Agent Skills by K-Dense

A library of instructions, examples and scripts for scientific work.
Compatible with Codex and other agents that support Agent Skills.

literature-review

Search, screen and synthesise published evidence.

citation-management

Check references and prepare citation records.

scientific-visualization

Plan figures and work with plotting libraries.

Read the SKILL.md, choose one relevant workflow and check its outputs.

Literature tools through MCP

MCP connects the assistant to a tool that can retrieve records.

BioMCP

Search biomedical papers and retrieve article records.
Sources include PubMed and Europe PMC.
Full-text availability depends on the source and access rights.

A local or hosted MCP server makes those tools available to the agent.
Follow the client setup guide and test a known paper first.

[Setup: biomcp.org/getting-started/mcp-clients/](https://biomcp.org/getting-started/mcp-clients/)

A literature task to try

A skill guides the workflow. An MCP tool retrieves the evidence.

“Find up to five primary studies on CD8 T-cell activation measured by flow cytometry in human samples. Use the connected literature tool and record the search. For each paper, report its DOI or PMID, study system, markers, sample size and a source passage. Mark fields you cannot verify and distinguish abstracts from full-text evidence. Save an evidence table.”

Check one row against the paper before using the table.